

CODE_DDP: EMOTIC B5-C3 Task-routed Adapter Bank

one frozen OpenAI CLIP · training-only prompt-free auxiliary route · deterministic class-introduction-task routing · fixed global inference scale

① Original DDP prompted prediction route

one image produces positive/negative class-specific token paths

Image input

$\mathbf{x}$

Class-specific positive/negative visual prompts

$$\mathbf{P}_V^c = \{\mathbf{P}_V^{c-}, \mathbf{P}_V^{c+}\}, \quad c \in \mathcal{C}^{0:t}$$

Frozen OpenAI CLIP encoders with DDP prompts

$$\mathbf{F}^{c\pm} = \mathbb{E}_V(\mathbf{P}_V^{c\pm}, \mathbf{x}) = [\mathbf{f}_0^{c\pm}, \dots, \mathbf{f}_N^{c\pm}],$$

$$\mathbf{t}^{c\pm} = \mathbb{E}_T(\mathbf{P}_T^{c\pm}, c), \quad \|\mathbf{t}^{c\pm}\|_2 = 1$$

Full token sequence

$$\mathbf{F}^{c\pm} = [\mathbf{f}_0^{c\pm}, \dots, \mathbf{f}_N^{c\pm}]$$

Prompted CLS representation

$$\mathbf{f}_{\text{cls}}^{c\pm} = \mathbf{f}_0^{c\pm}$$

Scaled token-text logits

$$q_n^{c\pm} = 20(\mathbf{t}^{c\pm})^\top \mathbf{f}_n^{c\pm}$$

Positive-path attention

$$\omega_n^c = \frac{\exp(q_n^{c+})}{\sum_{m=0}^N \exp(q_m^{c+})}$$

Shared-attention pooling

$$\bar{\mathbf{f}}^{c\pm} = \sum_{n=0}^N \omega_n^c \mathbf{f}_n^{c\pm}$$

Original DDP path logits

$$\ell_{\text{DDP}}^{c\pm} = 5 \sum_{n=0}^N \omega_n^c q_n^{c\pm} = 100(\mathbf{t}^{c\pm})^\top \bar{\mathbf{f}}^{c\pm}$$

Unchanged DDP logit pair

$$\ell_{\text{DDP}}^c = (\ell_{\text{DDP}}^{c-}, \ell_{\text{DDP}}^{c+})$$

② Task-routed Adapter Bank inference route

route every class—not the whole image—to its introduction-task Adapter

Class-to-task routing is fixed by the B5-C3 protocol

$$\tau(c) = s \iff c \in \mathcal{C}^s, \quad c \in \mathcal{C}^{0:t}$$

No sample-level task prediction · no oracle label · one multi-label image can invoke several Adapters

same frozen visual/text encoder parameters · different prompt usage and training role

Image → Bank

$$\mathbf{f}_{\text{cls}}^{c\pm}$$

Text → Bank

$$\mathbf{t}^{c\pm}$$

Frozen task Adapter cards accumulated over the stream

$$A_0 [\mathcal{C}^0]$$

$$A_1$$

...

$$A_t$$

Apply the Adapter assigned to class c

$$\mathbf{r}_{\tau(c)}^{c\pm} = W_{2,\tau(c)} \sigma \left(W_{1,\tau(c)} \frac{\mathbf{f}_{\text{cls}}^{c\pm}}{\|\mathbf{f}_{\text{cls}}^{c\pm}\|_2} \right)$$

Fixed Feature Difference at inference

$$\Delta \mathbf{f}^{c\pm} = \alpha \mathbf{r}_{\tau(c)}^{c\pm}, \quad \alpha = 0.03$$

Class-routed logit correction

$$\Delta \ell^{c\pm} = 100(\mathbf{t}^{c\pm})^\top \Delta \mathbf{f}^{c\pm}$$

③ Prompt-free CLIP auxiliary training route

TRAINING ONLY

enabled only while learning the current task Adapter; removed at inference

Image input

$\mathbf{x}$

Fixed current-task positive/negative prototypes

$$\mathbf{P}_{\text{proto}}^{c\pm}, \quad c \in \mathcal{C}^s$$

Same frozen OpenAI CLIP encoders without DDP prompts

$$\mathbf{z}_g = \text{norm}(\mathbb{E}_V(\mathbf{x})_{\text{cls}}),$$

$$\mathbf{u}^{c\pm} = \text{norm}(\mathbb{E}_T(\mathbf{P}_{\text{proto}}^{c\pm}))$$

Current task data and partial-label supervision

$$\mathcal{D}_s^{\text{Full}} = \{i : \sum_{c \in \mathcal{C}^s} y_{ic} > 0\} \quad \text{or} \quad K = 16 \text{ anchors/class},$$

$$M_{ic}^s = 1 \iff c \in \mathcal{C}^s$$

Train only the new task Adapter; old/future labels are masked

$$\mathbf{z}'_g = \text{norm}(\mathbf{z}_g + \gamma W_{2,s} \sigma(W_{1,s} \mathbf{z}_g)), \quad \gamma = 0.1$$

Current-task prototype logits

$$h_i^c = \kappa [\cos(\mathbf{z}'_{g,i}, \mathbf{u}^{c+}) - \cos(\mathbf{z}'_{g,i}, \mathbf{u}^{c-})], \quad c \in \mathcal{C}^s$$

Masked multi-label objective and identity preservation

$$\mathcal{L}_s = \text{BCE}_{M^*}(\mathbf{h}, \mathbf{y}) + \mu [1 - \cos(\mathbf{z}'_g, \mathbf{z}_g)]$$

Freeze the learned mapping and insert one new card

$$(W_{1,s}^*, W_{2,s}^*) = \arg \min \mathcal{L}_s, \quad A_s \xrightarrow{\text{freeze}} \{A_0, \dots, A_s\}$$

Strict incremental constraints

$$A_0, \dots, A_{s-1} \text{ frozen; } y^{\text{old/future}} \text{ unused; } \mathcal{D}_{\text{test}} \text{ unseen}$$

Independent final prediction: add the class-routed correction before the original DDP softmax

$$\tilde{\ell}^{c\pm} = \ell_{\text{DDP}}^{c\pm} + \Delta \ell^{c\pm}, \quad c \in \mathcal{C}^{0:t},$$

$$\hat{y}_{c+}^t = \frac{\exp(\tilde{\ell}^{c+}/\tau(t))}{\exp(\tilde{\ell}^{c-}/\tau(t)) + \exp(\tilde{\ell}^{c+}/\tau(t))},$$

one image $\longrightarrow \{A_{\tau(c)} : c \in \mathcal{C}^{0:t}\}$ (several Adapters may be used in one forward pass)

Full Task-routed Adapter Bank · Final Test mAP 32.5615 ± 0.0280 · fixed global $\alpha = 0.03$ · no class-specific gate

original DDP token-attention path

frozen task-routed Adapter Bank

training-only prompt-free route

shared frozen CLIP parameters

independent final logit addition